pmf: $P(X=x) = f(x)$

discrete: $f(x) \geq 0$, $\sum_x f(x) = 1$

cdf: $F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$, $-\infty < x < \infty$

discrete: $P(a \leq x \leq b) = F(b) - F(a)$

For continuous r.v., $P(X = a) = 0$ for any a

pdf: cont.

① $f(x) \geq 0$ for all $x \in \mathbb{R}$

② $\int_{-\infty}^{\infty} f(x)\,dx = 1$

③ $P(a < X < b) = \int_a^b f(x)\,dx$

cdf: cont. for a function w/ pdf $f(x)$,

$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx$

$P(a \leq x \leq b) = F(b) - F(a)$

$f(x) = \frac{dF(x)}{dx}$

joint pmf: discrete r.v.'s X and Y,

① $f(x,y) \geq 0$ for all (x,y)

② $\sum_x \sum_y f(x,y) = 1$

③ $P[X, Y) \in A] = \sum\sum_A f(x,y)$

for any region A in xy-plane,

marginal pmf: discrete

X: $g(x) = \sum_y f(x,y)$

Y: $h(y) = \sum_x f(x,y)$

conditional distribution: cont. and discrete

$f(Y|X) = \frac{f(x,y)}{g(x)}$, $g(x) > 0$

$f(X|Y) = \frac{f(x,y)}{h(y)}$, $h(y) > 0$

joint pdf:

continuous r.v.'s X and Y,

① $f(x,y) \geq 0$ for all (x,y)

② $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

③ $P[(X,Y) \in A] = \iint_A f(x,y)\,dx\,dy$

volume under surface above xy-plane

X and Y are statistically independent

$\Leftrightarrow f(x,y) = g(x)h(y) \Rightarrow f(x|y) = g(x)$

$X_1, \ldots X_n$ mutually independent

$\Leftrightarrow f(x_1, \ldots x_n) = f_1(x_1) f_2(x_2) \ldots f_n(x_n)$

Expectation:

$M = E(X) = \begin{cases} \sum_x x f(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} x f(x)\,dx, & X \text{ cont.} \end{cases}$

$E(g(X)) = \begin{cases} \sum_x g(x) f(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x) f(x)\,dx, & X \text{ cont.} \end{cases}$

Variance: ≥ 0 always

$\text{var}(X) = \sigma^2 = E[(X-M)^2] = E(X^2) - M^2$

$\sigma^2_{g(X)} = E\{[g(X) - M_{g(X)}]^2\} = E[g^2(X)] - E^2[g(X)]$

$\text{var}(aX+b) = a^2\text{var}(X)$, $\text{var}(a) = 0$, $\text{var}(X) = 0 \Leftrightarrow X = E(X)$

covariance: not scale-free $\Rightarrow$ magnitude $\neq$ strength b/w X, Y

$\sigma_{XY} = \text{cov}(X,Y) = E[(X - E(X))(Y - E(Y))]$

$= E(XY) - E(X)E(Y)$

$\text{cov}(X,Y) = 0$ if X and Y are independent r.v.'s.

$\text{cov}(aX+b, cY+d) = ac\,\text{cov}(X,Y)$

$\text{cov}(X+Y, Z) = \text{cov}(X,Z) + \text{cov}(Y,Z)$

correlation coefficient: $\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$

↑ scale-free; 1 when $Y = aX + b$

$P(aX+b, cY+d) = P(X,Y)$

memoryless property: $P(X > s \mid X > t) = P(X > s-t)$

prob of getting another $s-t$ failures after t failures is same as prob. of getting $s-t$ failures from the start.

Made with Goodnotes

Discrete uniform:

$f(x,k) = \frac{1}{k}$

$E(X) = \frac{1}{k}\sum_{i=1}^{k} x_i$

$\text{var}(X) = \frac{1}{k}\sum_{i=1}^{k}(x_i - M)^2$

Bernoulli: 2 outcomes

$X = \begin{cases} 1, & \text{success} \\ 0, & \text{fail} \end{cases}$

$f(1) = p$, $f(0) = q$

Binomial: number of successes in n independent Bernoulli trials

X

$b(x; n, p) = f(x) = \binom{n}{x} p^x q^{n-x}$

$\sum_x f(x) = \binom{n}{0}p^0 q^n + \binom{n}{1}p^1 q^{n-1} + \ldots + \binom{n}{n}p^n q^0 = 1$

$M = nE[I_i] = np$ — n independent Bernoulli variables: $X = I_1 + I_2 + \ldots + I_n$

$\text{var}(X) = \text{var}(I_1) + \text{var}(I_2) + \ldots + \text{var}(I_n) = npq$

Multinomial: a trial can result in k outcomes $E_1, \ldots E_k$ probabilities $p_1, p_2, \ldots p_k$. Then prob dist. of r.v.'s $X_1, \ldots X_k$ representing number of successes for $E_1, \ldots E_k$ in n ind. trials:

$f(x_1, \ldots x_k; p_1, \ldots p_k, n) = \binom{n}{x_1, x_2, \ldots x_k} p_1^{x_1} p_2^{x_2} \ldots p_k^{x_k}$

with $\sum x_i = n$, $\sum p_k = 1$ — $\binom{n}{x_1}\binom{n-x_1}{x_2}\binom{n-x_1-x_2}{x_3}\ldots\binom{x_k}{x_k}$

Hypergeometric: sampling w/o replacement $\Rightarrow$ trials are dep.

number of successes in random sample of size n out of N items with k successes, $N-k$ failure:

$h(x; N, n, k) = f(x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$

$M = \frac{nk}{N}$

$\sigma^2 = \frac{N-n}{N-1} n \frac{k}{N}\left(1 - \frac{k}{N}\right)$

$\frac{n}{N} \leq 0.05$

approx with binomial if $n \ll N$, use $p = \frac{k}{N}$

negative binomial dist: prob that k^{th} success is on y^{th} trial

k const.

$b^*(x; k, p) = f(x) = \binom{x-1}{k-1} p^k q^{x-k}$, $x = k, k+1, \ldots$

$M = \frac{k}{p}$, $\sigma^2 = \frac{k(1-p)}{p^2}$

geometric dist: number of trials where 1st success occurs in a sequence of Bernoulli trials $\Rightarrow$ negative binomial with $k = 1$

$g(x; p) = pq^{x-1}$, $x = 1, 2, 3, \ldots$

$M = \frac{1}{p}$, $\sigma^2 = \frac{1-p}{p^2}$

Poisson distribution: pmf of X representing number of outcomes in given interval t is:

$p(x; \lambda t) = f(x) = \frac{e^{-\lambda t}(\lambda t)^x}{x!}$, $x = 0, 1, 2, \ldots$

where λ is avg. number of outcomes per unit time/dist...

$M = \sigma^2 = \lambda t$

① $N(0) = 0$ ② independent increments ③ stationary increments

④ no simultaneous outcomes

prob. dist only dep. on length of interval

Poisson is approx. binomial for rare occur.

$X \sim b(x; n, p)$, when $n \to \infty$, $p \to 0$, good for $n \geq 20$, $p \leq 0.05$ or $n \geq 100$, $np \leq 10$

$np \xrightarrow[n\to\infty]{} \mu$ const, then $b(x; n, p) \to p(x, \mu)$

reliability: prob a product will function properly for at least a specified time under specified conditions

$R(t) = P(T > t) = \int_t^{\infty} f(t)\,dt = 1 - F(t)$

time until failure; pdf; cdf

conditional prob that product fails between $T = t$ to $T = t + \Delta t$, given it survived until t:

$\frac{P(t < T < t+\Delta t)}{P(t < T)} = \frac{F(t+\Delta t) - F(t)}{R(t)}$

failure rate: change over time of conditional prob

$z(t) = \lim_{\Delta t \to 0} \frac{\frac{F(t+\Delta t) - F(t)}{R(t)}}{\Delta t}$

$= \frac{F'(t)}{R(t)} = \frac{f(t)}{R(t)} = \frac{f(t)}{1 - F(t)}$

Thm: $X_1, \ldots X_n$ ind. χ^2 r.v.'s w/ dist. $\nu_1, \ldots \nu_n$

$Y = X_1 + \ldots + X_n \sim \chi^2(\nu_1 + \ldots + \nu_n)$

corollary: $X_1 \ldots X_n$ iid $N(M, \sigma^2) \Rightarrow Y = \sum_{i=1}^{n}\left(\frac{x_i - M}{\sigma}\right)^2 \sim \chi^2(n)$

continuous uniform:

$f(x; A, B) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B \\ 0, & \text{otherwise} \end{cases}$

$M = \frac{A+B}{2}$, $\sigma^2 = \frac{(B-A)^2}{12}$

normal distribution:

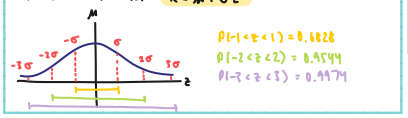
$n(x; M, \sigma) = f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{1}{2\sigma^2}(x-M)^2}$

standardization, z-scores

normal r.v. $X \to$ normal r.v. Z w/ $M = 0$, $\sigma^2 = 1$

$Z = \frac{X - M}{\sigma}$, $P(x_1 \leq X \leq x_2) = P(z_1 \leq Z \leq z_2)$

z score tells how many std. original observation falls from mean $x = M + \sigma z$



$P(-1 < z < 1) = 0.6828$

$P(-2 < z < 2) = 0.9544$

$P(-3 < z < 3) = 0.9974$

percentile: $P(Z < z) = p \Rightarrow P(X < x) = p$

p^{th} percentile

gamma function: $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x}\,dx$, $\alpha > 0$

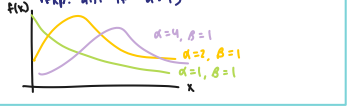
recursive: $\Gamma(\alpha) = (\alpha-1)\,\Gamma(\alpha-1)$, $\Gamma(n) = (n-1)!$, n integer

$\Gamma(\frac{1}{2}) = \sqrt{\pi}$ $\Gamma(1) = 1$

cont. r.v. X has gamma dist. w/ params α, β

$f(x; \alpha, \beta) = f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}$, $x > 0$

where $\alpha, \beta > 0$. $M = \alpha\beta$, $\sigma^2 = \alpha\beta^2$



exp dist: $f(x) = \frac{1}{\beta} e^{-x/\beta}$, $x > 0$

memoryless: $P(X > t) = P(X > t_0 + t \mid X > t_0)$

Weibull dist:

$f(x; \alpha, \beta) = \begin{cases} \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$

$F(x) = 1 - e^{-\alpha x^\beta}$, $x > 0$

$M = \alpha^{-1/\beta}\,\Gamma(1 + \frac{1}{\beta})$

$\sigma^2 = \alpha^{-2/\beta}\{\Gamma(1+\frac{2}{\beta}) - [\Gamma(1+\frac{1}{\beta})]^2\}$

failure rate at time t: $z(t) = \alpha\beta x^{\beta-1}$, $t > 0$

if $\beta = 1$: $z(t) = \alpha$

if $\beta > 1$: $z(t)$ increasing fcn. of t (wears)

if $\beta < 1$: $z(t)$ decreasing fcn. of t (strengthens)

chi-squared dist: special case of gamma dist

$\alpha = \frac{r}{2}$, $\beta = 2 \Rightarrow M = r$, $\sigma^2 = 2r$

normal approximation of binomial $np \geq 5$, $nq \geq 5$

X is binomial r.v. with $M = np$, $\sigma^2 = npq$

standard normal: $z = \frac{x - np}{\sqrt{npq}}$ as $n \to \infty$

$P(X \leq x) = \sum_{k=0}^{x} b(k; n, p) \approx P\left(z \leq \frac{x + 0.5 - np}{\sqrt{npq}}\right)$

discrete r.v. w/ pmf $f(x)$

$Y = U(X)$, $X = w(y)$ is 1-to-1

$g(y) = f(w(y))$

discrete r.v.'s with joint pmf $f(x_1, x_2)$

$X_1 = W_1(y_1, y_2)$, $X_2 = W_2(y_1, y_2)$ 1-to-1

$g(y_1, y_2) = f[w_1(x_1, x_2), w_2(x_1, x_2)]$

cont. r.v. w/ pdf $f(x)$

$Y = U(X)$, $X = w(y)$ is 1-to-1

$g(y) = f[w(y)]\,|J|$

$J = w'(y)$

$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$

cont. r.v.'s with joint pdf $f(x_1, x_2)$

$X_1 = W_1(y_1, y_2)$, $X_2 = W_2(y_1, y_2)$, 1-to-1

$g(y_1, y_2) = f[w_1(y_1, y_2), w_2(y_1, y_2)]\,|J|$

cont r.v. w/ pdf $f(x)$

$u = Y(x)$ not 1-to-1 $\Rightarrow$ partition k interval

$X_1 = W_1(y)$, $X_2 = W_2(y)$, ... $X_k = W_k(y)$

with 1-to-1 correspondance.

$g(y) = \sum_{i=1}^{k} f[w_i(y)]\,J_i$, $J_i = w_i'(y)$

r^{th} moment about zero of r.v. X:

$M_r' = E(X^r) = \begin{cases} \sum x^r f(x), & X \text{ disc.} \\ \int_{-\infty}^{\infty} x^r f(x)\,dx, & X \text{ cont.} \end{cases}$

first moment: $E(X) = M$

second moment: $E(X^2) = \sigma^2 + M^2$

central moment (moment around M)

$E[(X-M)^r]$, $E[(X-M)^2] = \sigma^2$

mgf of X

$M_X(t) = E(e^{tx}) = \begin{cases} \sum_x e^{tx} f(x), & \text{disc.} \\ \int_{-\infty}^{\infty} e^{tx} f(x), & \text{cont.} \end{cases}$

Thm: $\left.\frac{d^r M_X(t)}{dt^r}\right|_{t=0} = M_r'$

$M = \left.\frac{dM_X}{dt}\right|_{t=0}$ $\sigma^2 = \left.\frac{d^2 M_X(t)}{dt^2}\right|_{t=0} - M^2$

Thm: X and Y are r.v.'s w/ mgf's $M_X(t)$, $M_Y(t)$

If $M_X(t) = M_Y(t)$ for all t, then X and Y have same dist.

① $Y = aX + b \Rightarrow M_Y(t) = e^{at} M_X(bt)$

② $X_1, \ldots X_n$ ind. r.v.'s w/ mgf's $M_{X_1}(t), \ldots M_{X_n}(t)$

and $Y = X_1 + \ldots + X_n \Rightarrow M_Y(t) = M_{X_1}(t)\,M_{X_2}(t) \ldots M_{X_n}(t)$

Thm: $X \sim N(M, \sigma^2) \Rightarrow M_X(t) = \exp(Mt + \frac{1}{2}\sigma^2 t^2)$

$X \sim \chi^2(\nu) \Rightarrow M_X(t) = (1-2t)^{-\nu/2}$

Thm: $X_1, \ldots X_n$ ind. r.v.'s w/ normal dist (M_i, σ_i^2)

$Y = a_1X_1 + \ldots + a_nX_n \sim N(a_1M_1 + \ldots + a_nM_n, a_1^2\sigma_1^2 + \ldots + a_n^2\sigma_n^2)$

① $X_1 + X_2 + \ldots + X_n \sim N(M_1 + \ldots + M_n, \sigma_1^2 + \ldots + \sigma_n^2)$

② $X_1, \ldots X_n \overset{iid}{\sim} N(M, \sigma^2) \Rightarrow Y = X_1 + \ldots + X_n \sim N(nM, n\sigma^2)$

iid = independently identically dist. $\bar{Y} = \frac{1}{n}(X_1 + \ldots + X_n) \sim N(M, \frac{\sigma^2}{n})$

